

I have done more literature review and have boiled the model down to three candidates. Our requirements are as follows:

- Spatiotemporal model (in 1D for now)
- Isotropic in Space
- Non-Separable in any dimension

Ideal Conditions are:

- Matern in space
- Matern/diffuse in time

The base spatial model is as follows $(\kappa^2 - \Delta)^{\alpha/2} Z(h) = W(h)$ where Δ is the laplacian, κ is the range parameter, α is the smoothness paramter, h is our coordinate (space), Z is our process, and W is a white noise process

The first candidate for a spatiotemporal model is extending the laplacian to time

$$\Delta = \frac{\partial^2}{\partial t^2} + \sum_{x_i \in x} \frac{\partial^2}{\partial x_i^2}$$

This does not fit our criteria but is good

Next is one where we add a first partial of time in the linear operator, $L = \kappa^2 + \Delta_s + \partial_t$ and then take $L^{\alpha/2} Z(h, u) = W(h, u)$ While I think this is a good start, I am doubtful on the flexibility of the model even when a sufficient amount of parameters are added.

Another Candiadate is the model defined in the Paper <https://www.stat.purdue.edu/tlzhang/maternasy> Where we consider the NFSST Matern Model

$$M_{r,a_1,a_2,\nu_1,\nu_2}(h, u) = E \left[\exp \left(-\frac{1}{2} \left(\frac{a_1^2 \|h\|^2}{V_1} + \frac{2a_1 a_2 u r^T h}{\sqrt{V_1 V_2}} + \frac{a_2^2 u^2}{V_2} \right) \right) \right]$$

Where the h is the spacial coordinate, u is the temporal coordinate.

M is called the NFSST (Not fully symmetric space time) Matern Correlation function.

I really like this model. Since it fully fits every single requirement and ideal condition. Also I can see a great next step being a SPDE based FEM solver since I am not able to find much done on this specific topic.

Next steps are as follows

- Review the Paper further, especially what the interaction parameter is and how to estimate it.
- Understand the fourier transform of this model for the frequency behavior and then the PIT of the model both in space and time.
- Develop a linear operator L s.t. it has the same spectral density as the model.
- Develop a method to solve the SPDE numerically using some FEM method and potentially adapting the R-INLA or RSPDE libraries solver.